

A Photocount Reading of the Interpolating Function, and a No-Go for Deriving Its Normalisation from a Clock

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Abstract

A companion paper removed the MOND acceleration scale as an independent parameter and found that 155 SPARC rotation curves then select, with nothing fitted, the member $n = 2$ of the family $\mu_n(Y) = 1 - (1 + Y)^{-n}$, which is $\kappa = 1/2$. This paper reports what happened when we tried to derive that integer, and one structural result obtained along the way. **First, a no-go.** A cuscuton clock's own field equation fixes $V'(\tau) = -3HU$, so a vanishing potential forces a vanishing expansion rate: any cuscuton clock in an expanding universe *requires* a potential. A potential is precisely the free additive constant that makes the acceleration-scale normalisation a zero mode. Hence the twelve-gate relativistic construction of our earlier work provably cannot derive κ , and the two tracks cannot be merged — they assign values fourteen orders apart to the same coefficient, and forcing agreement demands a margin exceeding one. **Second, an identification.** The family is exactly Mandel's n -mode photocount formula: for n independent thermal modes of mean occupancy Y , μ_n is the probability that at least one quantum is present. Equivalently it is one minus a Tsallis q -exponential with $q = 1 + 1/n$, the ordinary exponential kernel being the infinite-mode limit. This explains why the exponent is an *integer* — it is a mode count — without fixing its value, and replaces the question with a sharper one: why does the vacuum present exactly two modes to an acceleration? **Third, the answer to that is not available, and we report the closures.** The modes are not Unruh modes: a fixed-frequency thermal occupancy is exponentially small where the reading requires it linear, failing by 41 orders at $Y = 10^{-2}$. Twelve pre-registered mathematical criteria scatter over three different members, with $n = 1$ the modal choice; only the convergence boundary selects $n = 2$, and as a boundary rather than an optimum. Effective-field-theory positivity is inapplicable (the kinetic matrix degenerates where $\mu(0) = 0$) and causality is saturated identically by every kernel. No published entropic or holographic derivation produces the family beyond $n = 1$. Finally, the d -dimensional generalisation $\mu_n^{(d)} = 1 - (1 + Y^{d-2})^{-n}$ exists and is forced, but the mode count enters the deep limit as a multiplicative prefactor and cannot change a power — so it is dimensionally inert and no dimension fixes it. **We conclude that the exponent is an empirical quantity, and that this is the result rather than a placeholder for one.**

1 A no-go for the clock

Our earlier relativistic construction [1] carries a cuscuton clock τ with action $\int \sqrt{-g} [s W(Y, \tau) - V(\tau) + \dots]$, $s = \sqrt{-\nabla\tau \cdot \nabla\tau}$. Varying τ at constant $U \equiv W(0, \tau)$ gives $U \nabla_\mu n^\mu = -V'(\tau)$, and in a Friedmann background $\nabla_\mu n^\mu = 3H$, so

$$V'(\tau) = -3HU. \tag{1}$$

A vanishing potential therefore forces $H = 0$: **any cusciton clock in an expanding universe requires a potential.**

This matters because of a zero mode. For any *free* interpolating function F of derivative invariants, the shift $F \rightarrow F + c$ leaves F' untouched — and F' is all the scalar equation and the static force law ever see — while shifting the metric-multiplying piece by exactly c , which is a cosmological constant and nothing else. The normalisation relating a_0 to the vacuum energy is therefore unfixable whenever the action contains a free additive constant, *regardless* of field content [2]. Equation (1) says the clock construction cannot dispense with one. It is therefore inside the class the no-go covers, and no further work on it will produce κ .

The two tracks also cannot be merged directly. Both call $W(0)$ the same coefficient U ; the clock construction gives $U = m_{\text{rel}}\rho_{\text{dm}} = 3.68 \times 10^{-25} \text{ eV}^4$ at its parameter point, while the parameter-free reading gives $U = \rho_{\Lambda} = 2.52 \times 10^{-11} \text{ eV}^4$, a ratio of 6.9×10^{13} . Forcing agreement needs $m_{\text{rel}} = \rho_{\Lambda}/\rho_{\text{dm}} = 2.58$, and $m_{\text{rel}} = 1 - \mu$ cannot exceed unity. The merge is arithmetically blocked, not merely strained. *The one loophole:* (1) is written at constant U ; with U depending on τ there is an extra term, and whether it can substitute for the potential is not computed here.

An architecture that is both relativistic and κ -deriving must therefore have its expansion driven by the *same* function that carries the gradient sector, with no independent potential. That rules out the cusciton as the timekeeper.

2 The family is a photocount formula

A single bosonic mode in thermal equilibrium with mean occupancy Y has the geometric distribution $P(k) = Y^k/(1+Y)^{k+1}$, so it is empty with probability $1/(1+Y)$. For n independent such modes,

$$\mu_n(Y) = 1 - (1+Y)^{-n} = P(\text{at least one quantum present}), \quad (2)$$

which is Mandel's n -mode photocount formula. We verified the identification symbolically and by Monte Carlo (400,000 trials at $Y = 2$, $n = 2$: 0.88914 against the closed form 0.88889). Under the principle of [2], Y is the acceleration in units of the dark-energy acceleration, so the reading is direct: *the response to an acceleration is the probability that it excites at least one quantum out of the vacuum.*

The same family is one minus a Tsallis q -exponential, $\mu_n = 1 - \exp_q(-nY)$ with $q = 1 + 1/n$, so the mode count and the deformation parameter are one quantity, and the exponential kernel in common use is the infinite-mode limit — the case in which the mode count has been forgotten. The composition law $1 - \mu_n = (1 - \mu_1)^n$ becomes obvious: n independent modes.

This buys the *form* and not the *value*. It explains why the exponent is an integer, and replaces the question with: **why does the vacuum present exactly two independent modes to an acceleration?**

3 The modes are not Unruh modes

The obvious answer is that an accelerating body sees a bath at $T = g/2\pi$ and the modes are that bath's. A mode of frequency $\omega = s/2\pi$ does give occupancy g/s — but only in the classical, high-occupancy limit. The photocount reading needs occupancy *exactly* $Y = g/s$ at all accelerations,

and a thermal occupancy is exponentially small at low temperature:

$Y = g/s$	reading requires	thermal gives
1	1	0.582
10^{-1}	10^{-1}	4.5×10^{-5}
10^{-2}	10^{-2}	3.7×10^{-44}

Forty-one orders, in the only regime MOND is about. The modes are not Unruh modes, nor any fixed-frequency thermal modes. This matches independently the fact that every thermal-screen derivation in the literature has *exponentially small* deep-MOND corrections where this family's are analytic. What survives is that the occupancy is exactly linear in the acceleration — an equipartition statement, how many quanta of the dark-energy acceleration fit into this one, not a Boltzmann one.

4 Four closed routes

Pre-registered mathematical criteria. Twelve natural distinguishing properties of the family were fixed before evaluation, each statable without reference to the number two. Nine gave a unique selection, scattering over three members: $n = 1$ takes five, $n = 3/2$ two, $n = 2$ two. The modal choice is $n = 1$. The two criteria selecting $n = 2$ — convergence of $\int(1 - \mu_n)$ and finiteness of $\lim(F - z)$ — are the same fact in two dressings, and select it as a *boundary* rather than an interior optimum. With nine criteria over three members, chance places three on any one. The one substantive residue is that $n = 2$ is the *marginal* index, which is what a criticality argument would want; that is recorded as suggestive and not more.

Effective field theory. Positivity bounds are inapplicable: with $\mu(0) = 0$ the fluctuation kinetic matrix degenerates at zero gradient, so there is no propagator and no S -matrix, and AQUAL is not a derivative expansion. Granting them anyway, the single scale-free invariant is $9(n+1)/(16n)$, continuous and monotone, so a two-sided bound could exclude a range but never select an integer. Causality is non-discriminating: around a static gradient $c_{\parallel}^2 = 1 + d \ln \mu / d \ln Y$, which tends to 2 in the deep regime for *every* member of the family, and indeed for any kernel with a deep-MOND limit, since $\mu \sim Y^p$ gives $c_{\parallel}^2 \rightarrow p + 1$ and MOND forces $p = 1$. Nobody has aimed positivity or causality at MOND interpolating functions before.

Existing derivations. No published entropic or holographic derivation produces this family beyond the $n = 1$ member, which modified-Rényi entropic gravity gives exactly. Every factor of two in that literature sits in the acceleration scale, never in the exponent.

Dimension. Keeping the photocount structure, the dimension can only enter through the occupancy, giving $\mu_n^{(d)}(Y) = 1 - (1 + Y^{d-2})^{-n}$, which reduces to the family at $d = 3$ and carries Milgrom's conformally invariant deep power in general. But in the deep limit the response is count $\times$ occupancy, so the count is a multiplicative prefactor and cannot change a power: $d \ln(\text{response}) / d \ln Y = d - 2$, free of n . The count is *dimensionally inert*, so no dimension fixes it. Four candidate counts — transverse directions, graviton polarisations, screen dimensions, and the two branches of the gradient invariant — all give two at $d = 3$ and the generalisation separates none of them. A by-product: the equipartition reading of the occupancy is specifically a three-dimensional fact.

5 Conclusion

The exponent is an empirical quantity. The rotation curves prefer $n = 2$ over $n = 1$ by 0.016 dex [2]; sharpening that means beating the relation's ~ 0.045 dex intrinsic scatter by a factor of a few, which needs better mass-to-light ratios and distances rather than better theory. After the closures above we do not regard this as a placeholder for a derivation that will appear shortly. **It is the result.**

6 What is not claimed

The clock no-go is written at constant U and the τ -dependent case is not computed. The photocount identification is an *identification*: nothing here shows the vacuum response *is* such a count, only that the function the data selected has that form; and the analytic-versus-exponential mismatch cuts against reading it as a literal thermal screen. Twelve criteria is not exhaustive. The d -dimensional deep power is Milgrom's conformal argument, quoted. Uniqueness of the generalisation holds within the photocount reading only. The Unruh test uses a single fixed frequency; a distribution of frequencies could give linear occupancy, but a Gamma-distributed rate is exactly what generates this family, so that route risks assuming the answer. This paper concerns one coefficient and is not a verdict on any construction.

7 Reproducibility

fable_independent_2026/ lanes L235-L239 with committed outputs and results files, and hunt_2026/eft01_positivity_causality_mu_family_2026.py. Lean file lean_2026/Mondlean.lean: 160 theorems, exit 0, zero sorry.

References

- [1] C. P. Zimmerman, *A Conserved Clock Rate, and the Board Run End to End*, doi:10.5281/zenodo.22731066 (2026).
- [2] C. P. Zimmerman, *A Parameter-Free Radial Acceleration Relation from the Dark-Energy Scale*, doi:10.5281/zenodo.22731370 (2026).
- [3] B. Famaey, S. McGaugh, Living Rev. Relativity **15**, 10 (2012), arXiv:1112.3960.
- [4] L. Mandel, E. Wolf, *Optical Coherence and Quantum Optics*, Cambridge (1995).
- [5] N. Afshordi, D. J. H. Chung, G. Geshnizjani, Phys. Rev. D **75**, 083513 (2007).
- [6] J.-P. Bruneton, G. Esposito-Farèse, Phys. Rev. D **76**, 124012 (2007), arXiv:0705.4043.
- [7] The research repository, <https://github.com/carlzimmerman/zimmerman-formula> (2026).